HW3

Zakari L. Billo

2026-04-02

Load Packages

```
library(caret)
library(tidymodels)
library(discrim)
library(MASS)
library(mlbench)
library(pROC)
library(klaR)
library(plotmo)
library(splines)
library(mgcv)
library(pdp)
library(earth)
library(tidyverse)
library(ggplot2)
library(ISLR)
library(glmnet)

knitr::opts_chunk$set(
  fig.width = 8,
  fig.height = 7,
  fig.align = "center",
  out.width = "90%"
)

theme_set(theme_minimal() + theme(legend.position = "bottom"))

options(
  ggplot2.continuous.colour = "viridis",
  ggplot2.continuous.fill = "viridis"
)

scale_colour_discrete = scale_colour_viridis_d
scale_fill_discrete = scale_fill_viridis_d
```

Split the dataset into two parts: training data (70%) and test data (30%).

```
dat <- read.csv("data/auto.csv",  na = c("NA", ".", "")) |>
  janitor::clean_names() |> na.omit() |>
  mutate(
    mpg_cat = as.factor(mpg_cat),
    origin = factor(origin,
                    levels = c(1, 2, 3),
```

```r
                    labels = c("American", "European", "Japanese")),
  )

dat <- dat[dat$cylinders != 3, ]

summary(dat)
```

```
##    cylinders      displacement     horsepower        weight      acceleration
##  Min.   :4.000   Min.   : 68.0   Min.   : 46.0   Min.   :1613   Min.   : 8.00
##  1st Qu.:4.000   1st Qu.:105.0   1st Qu.: 75.0   1st Qu.:2225   1st Qu.:13.90
##  Median :4.000   Median :151.0   Median : 92.5   Median :2822   Median :15.50
##  Mean   :5.497   Mean   :195.7   Mean   :104.5   Mean   :2984   Mean   :15.56
##  3rd Qu.:8.000   3rd Qu.:302.0   3rd Qu.:129.0   3rd Qu.:3622   3rd Qu.:17.12
##  Max.   :8.000   Max.   :455.0   Max.   :230.0   Max.   :5140   Max.   :24.80
##       year            origin    mpg_cat
##  Min.   :70.00   American:245   high:195
##  1st Qu.:73.00   European: 68   low :193
##  Median :76.00   Japanese: 75
##  Mean   :75.98
##  3rd Qu.:79.00
##  Max.   :82.00
```

```r
set.seed(1)
data_split <- initial_split(dat, prop = 0.7)

# Extract the training and test data
training_data <- training(data_split)
testing_data <- testing(data_split)

# set up cross validation control
ctrl <- trainControl(method = "cv", 10,
                     summaryFunction = twoClassSummary,
                     classProbs = TRUE)
```

(a) Perform logistic regression analysis. Are there redundant predictors in your model? If so, identify them. If there are none, please provide an explanation.

```r
set.seed(1)

lr.fit <- train(
                mpg_cat ~ .,
                data=training_data,
                method="glm",
                family="binomial",
                metric = "ROC",
                preProcess=c("center", "scale"),
                trControl=ctrl)

summary(lr.fit)
```

```
##
## Call:
## NULL
##
## Coefficients:
```

```
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.9181     0.3850   2.385  0.01708 *
## cylinders        0.4121     0.9284   0.444  0.65712
## displacement    -1.4203     1.7979  -0.790  0.42956
## horsepower       2.2279     1.1460   1.944  0.05189 .
## weight           4.5912     1.5109   3.039  0.00238 **
## acceleration     0.2248     0.4751   0.473  0.63615
## year            -1.7152     0.3943  -4.350 1.36e-05 ***
## originEuropean  -0.4152     0.3529  -1.177  0.23938
## originJapanese  -0.5572     0.3790  -1.470  0.14153
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 375.505  on 270  degrees of freedom
## Residual deviance:  97.381  on 262  degrees of freedom
## AIC: 115.38
##
## Number of Fisher Scoring iterations: 8
```

```r
print(lr.fit)
```

```
## Generalized Linear Model
##
## 271 samples
##   7 predictor
##   2 classes: 'high', 'low'
##
## Pre-processing: centered (8), scaled (8)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 245, 244, 244, 243, 244, 244, ...
## Resampling results:
##
##   ROC        Sens       Spec
##   0.9739313  0.9214286  0.9010989
```

The final, unpenalized logistic regression model was

$$\begin{aligned}\text{logit}(p) = 0.9181 &+ 0.9284 \cdot \text{cylinders}^* - 1.4203 \cdot \text{displacement}^* \\ &+ 2.2279 \cdot \text{horsepower}^* + 4.5912 \cdot \text{weight}^* + 0.2248 \cdot \text{acceleration}^* \\ &- 1.7152 \cdot \text{year}^* - 0.4152 \cdot I(\text{origin} = \text{European}) \\ &- 0.5572 \cdot I(\text{origin} = \text{Japanese})\end{aligned}$$

Next, I used LASSO to examine the redundant features.

```r
set.seed(1)
lasso.fit <- train(
                  mpg_cat ~ .,
                  training_data,
                  method = "glmnet",
                  preProcess = c("center", "scale"),
                  tuneGrid = expand.grid(alpha = 1,
                                         lambda = exp(seq(10, -5, length = 100))),
                  trControl = ctrl)
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not
## in the result set. ROC will be used instead.
```

```r
# Extract coefficients at the optimal lambda
lasso_coefs <- coef(lasso.fit$finalModel, s = lasso.fit$bestTune$lambda)
print(lasso_coefs)
```

```
## 9 x 1 sparse Matrix of class "dgCMatrix"
##                 s=0.007840248
## (Intercept)      0.676426725
## cylinders        0.322464723
## displacement     .
## horsepower       0.863374970
## weight           3.032306515
## acceleration     .
## year            -1.161395896
## originEuropean  -0.008911498
## originJapanese  -0.159638749
```

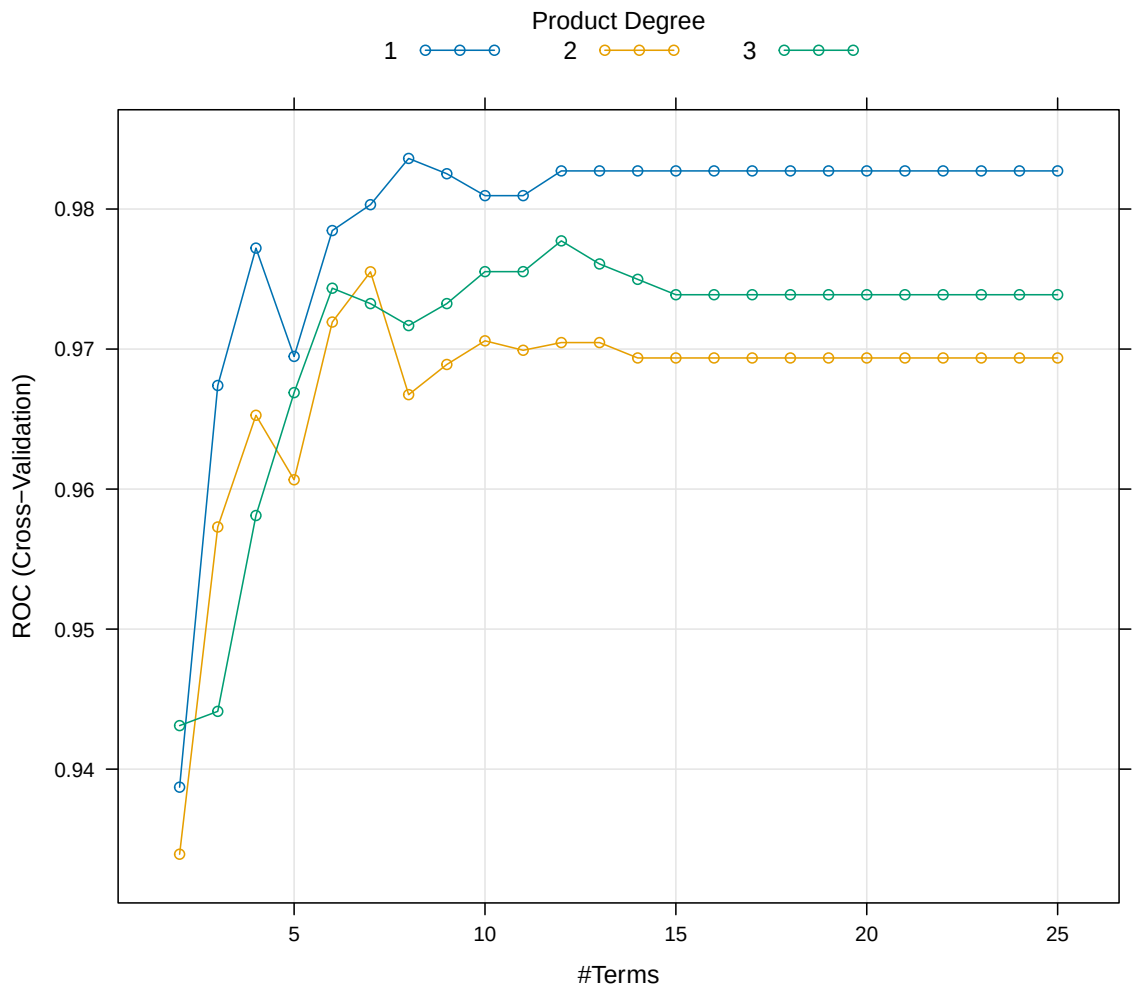
The redundant predictors (shrunk to 0 by the L1 norm of LASSO) appear to be `acceleration` and `displacement`. An explanation for this is these measures of engine performance add dimensionality but do not each describe unique properties of the car. For example, `acceleration` is the derivative of velocity (the variable `displacement` over time, in a direction), but both do not inform the mileage that a car may have.

The final penalized logistic regression model for LASSO was

$$\begin{aligned}\text{logit}(p) = 0.6764 &+ 0.3225 \cdot \text{cylinders}^* + 0.8634 \cdot \text{horsepower}^* \\ &+ 3.0323 \cdot \text{weight}^* - 1.1614 \cdot \text{year}^* \\ &- 0.0089 \cdot I(\text{origin} = \text{European}) - 0.1596 \cdot I(\text{origin} = \text{Japanese})\end{aligned}$$

(b) Train a multivariate adaptive regression spline (MARS) model. Does the MARS model improve prediction performance compared to logistic regression?

```r
set.seed(1)
mars_grid <- expand.grid(degree = 1:3, nprune = 2:25)

mars.fit <- train(mpg_cat ~ .,
                  data = training_data,
                   method = "earth",
                   tuneGrid = mars_grid,
                   metric = "ROC",
                   trControl = ctrl)

plot(mars.fit)
```




```r
mars.fit$bestTune
```

```
##   nprune degree
## 7      8      1
```

```r
coef(mars.fit$finalModel)
```

```
##         (Intercept) h(displacement-232)          h(year-72)      h(3525-weight)
##         8.802801446        -0.019110497        -0.581229595        -0.005436958
##      h(6-cylinders) h(displacement-156) h(displacement-122) h(displacement-119)
##        -2.045402852         0.174265469        -1.318868328         1.153811721
```

```r
print(mars.fit$finalModel)
```

```
## GLM (family binomial, link logit):
##  nulldev  df       dev  df   devratio     AIC iters converged
##  375.505 270   79.2733 263      0.789   95.27     7         1
##
## Earth selected 8 of 17 terms, and 4 of 8 predictors (nprune=8)
## Termination condition: Reached nk 21
## Importance: weight, year, cylinders, displacement, horsepower-unused, ...
## Number of terms at each degree of interaction: 1 7 (additive model)
## Earth GCV 0.06118465    RSS 14.7963    GRSq 0.7569021    RSq 0.7814586
```

```r
summary(mars.fit$finalModel)
```

```
## Call: earth(x=matrix[271,8], y=factor.object, keepxy=TRUE,
##             glm=list(family=function.object, maxit=100), degree=1, nprune=8)
##
## GLM coefficients
##                            low
## (Intercept)          8.8028014
## h(6-cylinders)      -2.0454029
## h(displacement-119)  1.1538117
## h(displacement-122) -1.3188683
## h(displacement-156)  0.1742655
## h(displacement-232) -0.0191105
## h(3525-weight)      -0.0054370
## h(year-72)          -0.5812296
##
## GLM (family binomial, link logit):
##  nulldev  df       dev  df   devratio     AIC iters converged
##  375.505 270   79.2733 263      0.789   95.27     7         1
##
## Earth selected 8 of 17 terms, and 4 of 8 predictors (nprune=8)
## Termination condition: Reached nk 21
## Importance: weight, year, cylinders, displacement, horsepower-unused, ...
## Number of terms at each degree of interaction: 1 7 (additive model)
## Earth GCV 0.06118465    RSS 14.7963    GRSq 0.7569021    RSq 0.7814586
```

The final mars model selected was

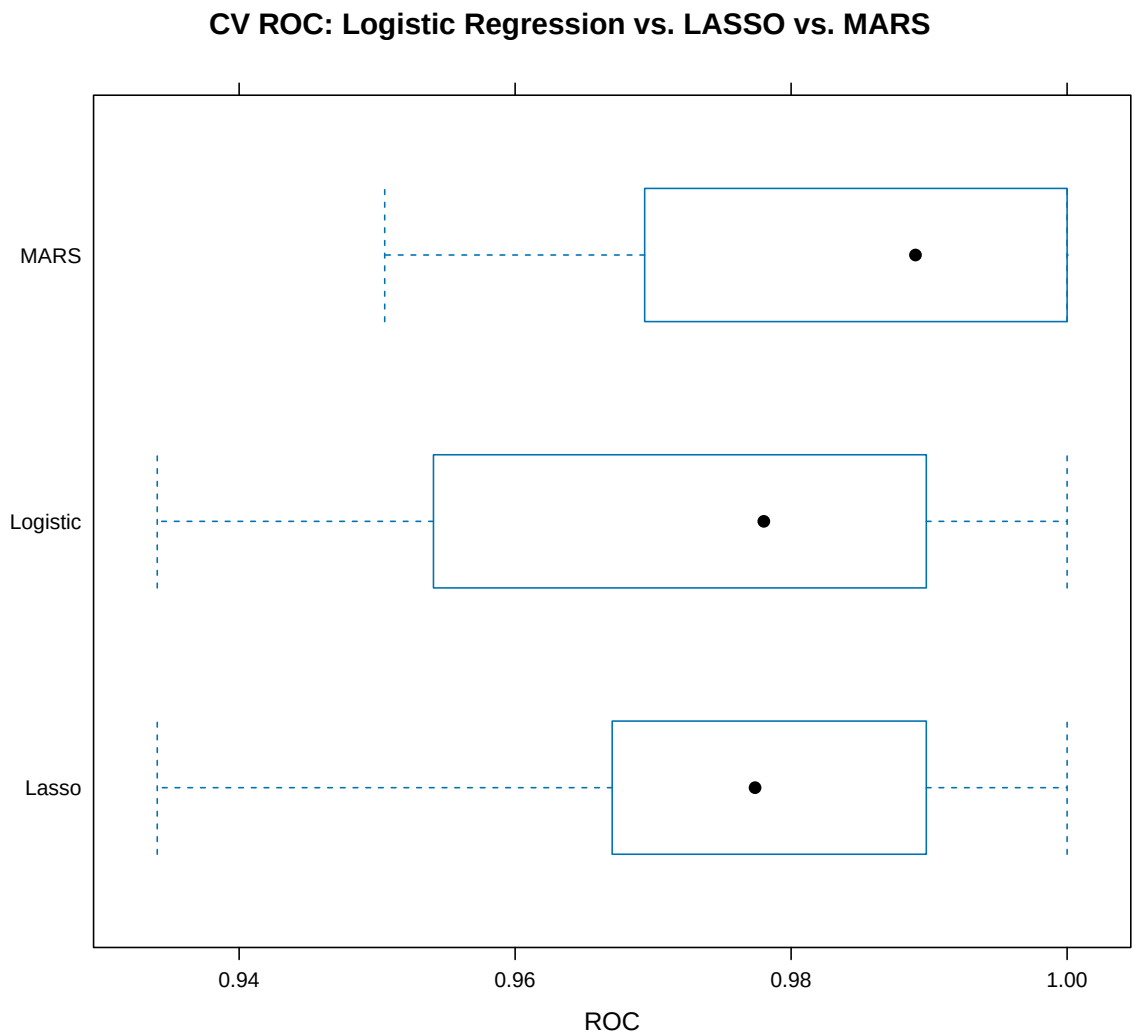
$$\begin{aligned}\text{logit}(p) = 8.8028 &- 2.0454 \cdot h(6 - \text{cylinders}) + 1.1538 \cdot h(\text{displacement} - 119) \\ &- 1.3189 \cdot h(\text{displacement} - 122) + 0.1743 \cdot h(\text{displacement} - 156) \\ &- 0.0191 \cdot h(\text{displacement} - 232) - 0.0054 \cdot h(3525 - \text{weight}) \\ &- 0.5812 \cdot h(\text{year} - 72)\end{aligned}$$

```r
set.seed(1)
res <- resamples(list(Logistic = lr.fit,
                      Lasso = lasso.fit,
                      MARS = mars.fit))

summary(res)
```

```
##
## Call:
## summary.resamples(object = res)
##
## Models: Logistic, Lasso, MARS
## Number of resamples: 10
##
## ROC
##               Min.   1st Qu.   Median      Mean   3rd Qu. Max. NA's
## Logistic 0.9340659 0.9600667 0.978022 0.9739313 0.9879091    1    0
## Lasso    0.9340659 0.9670330 0.977388 0.9766000 0.9895997    1    0
## MARS     0.9505495 0.9719388 0.989011 0.9836131 0.9992604    1    0
##
## Sens
```

```
##                Min.   1st Qu.    Median      Mean   3rd Qu. Max. NA's
## Logistic 0.7857143 0.8750000 0.9285714 0.9214286 0.9821429    1    0
## Lasso    0.8571429 0.8750000 0.9285714 0.9357143 1.0000000    1    0
## MARS     0.8571429 0.9285714 1.0000000 0.9642857 1.0000000    1    0
##
## Spec
##                Min.   1st Qu.    Median      Mean   3rd Qu. Max. NA's
## Logistic 0.7692308 0.9230769 0.9230769 0.9010989 0.9271978    1    0
## Lasso    0.7692308 0.8461538 0.9230769 0.8857143 0.9271978    1    0
## MARS     0.7692308 0.8653846 0.9230769 0.9159341 0.9821429    1    0
```

```r
bwplot(res, metric = "ROC", main = "CV ROC: Logistic Regression vs. LASSO vs. MARS")
```



Overall, it appears that the MARS model had a higher cross-validated ROC (0.984 vs. LASSO's 0.977 and LR's 0.974), Sensitivity (0.964 vs. LASSO's 0.936 and LR's 0.921) and Specificity (0.916 vs LASSO's 0.886 and LR's 0.901). Thus, the final cross-validated MARS model is an improvement over both the LASSO and Logistic regression model.

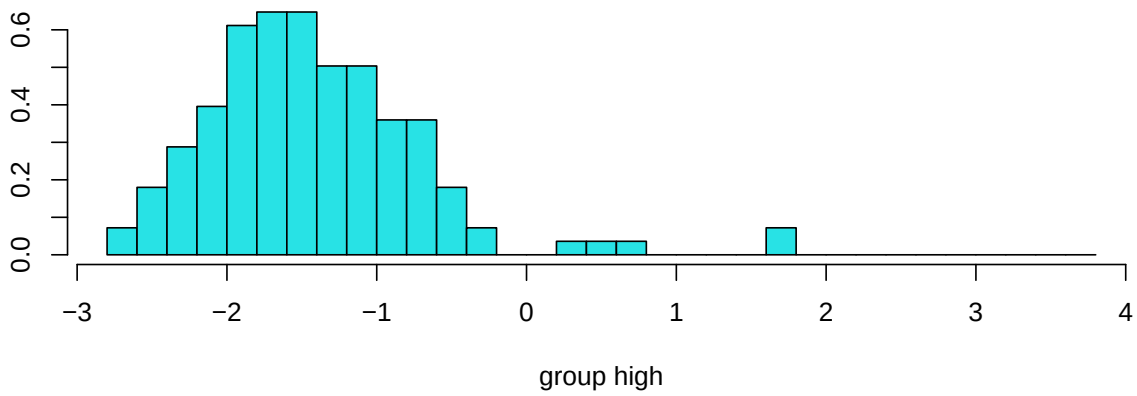
(c) Perform linear discriminant analysis using the training data. Plot the linear discriminant(s).

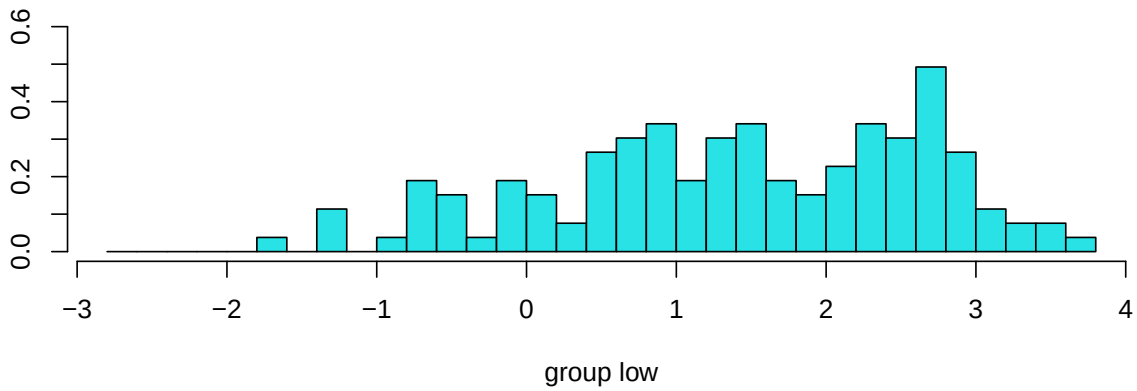
```r
# train LDA model with CV
# set up cross validation control
ctrl1 <- trainControl(method = "repeatedcv", repeats = 10,
```

```r
                      summaryFunction = twoClassSummary,
                      classProbs = TRUE)

set.seed(1)
lda.fit <- train(mpg_cat ~ .,
                   data = training_data,
                   method = "lda",
                   metric = "ROC",
                   trControl = ctrl)

lda.fit$finalModel
```

```
## Call:
## lda(x, grouping = y)
##
## Prior probabilities of groups:
##      high       low
## 0.5129151 0.4870849
##
## Group means:
##      cylinders displacement horsepower   weight acceleration     year
## high  4.165468     115.3885   77.83453 2312.525     16.54748 77.56835
## low   6.780303     273.3258  129.73485 3659.114     14.76591 74.33333
##      originEuropean originJapanese
## high     0.25179856     0.35971223
## low      0.07575758     0.03030303
##
## Coefficients of linear discriminants:
##                         LD1
## cylinders       0.516258992
## displacement   -0.001534288
## horsepower     -0.006862193
## weight          0.001161575
## acceleration    0.021341766
## year           -0.119127798
## originEuropean -0.154544602
## originJapanese -0.437282702
```

```r
# Fit a standalone LDA model just for the plot
lda.plot <- lda(mpg_cat ~ ., data = training_data)
plot(lda.plot)
```





The final LDA model was

$$\begin{aligned}\text{LD1} = 0.5163 \cdot \text{cylinders} - 0.0015 \cdot \text{displacement} - 0.0069 \cdot \text{horsepower} \\ + 0.0012 \cdot \text{weight} + 0.0213 \cdot \text{acceleration} - 0.1191 \cdot \text{year} \\ - 0.1545 \cdot I(\text{origin} = \text{European}) - 0.4373 \cdot I(\text{origin} = \text{Japanese})\end{aligned}$$

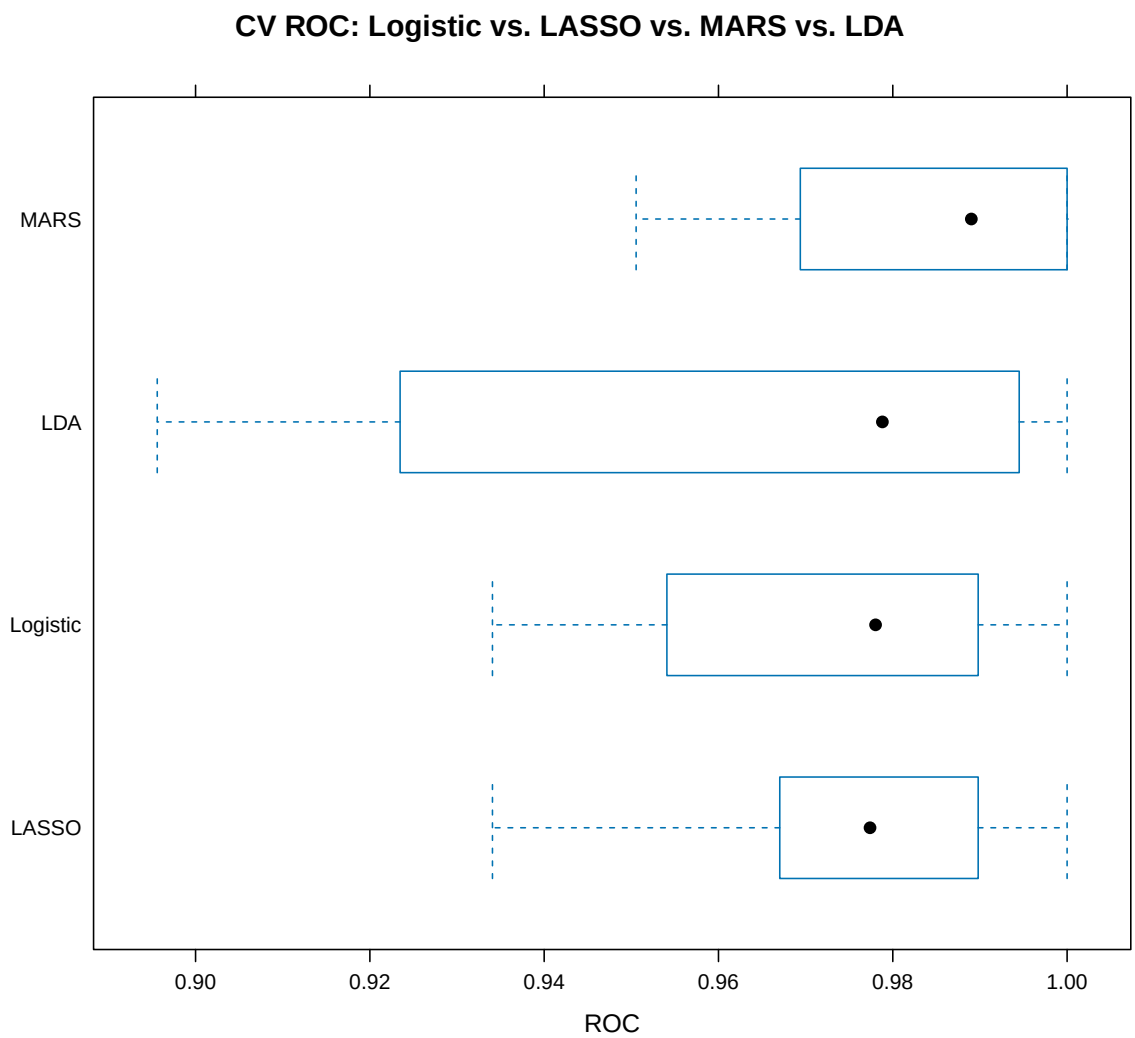
(d) Which model will you choose to predict the response variable? Plot its ROC curve and report the AUC.

```
set.seed(1)
res <- resamples(list(Logistic = lr.fit,
                      LASSO = lasso.fit,
                      MARS = mars.fit,
                      LDA = lda.fit))

summary(res)
```

```
##
## Call:
## summary.resamples(object = res)
##
## Models: Logistic, LASSO, MARS, LDA
## Number of resamples: 10
##
## ROC
##                Min.   1st Qu.    Median      Mean   3rd Qu. Max. NA's
```

```
## Logistic 0.9340659 0.9600667 0.9780220 0.9739313 0.9879091    1    0
## LASSO    0.9340659 0.9670330 0.9773880 0.9766000 0.9895997    1    0
## MARS     0.9505495 0.9719388 0.9890110 0.9836131 0.9992604    1    0
## LDA      0.8956044 0.9337263 0.9788069 0.9647778 0.9931319    1    0
##
## Sens
##               Min.   1st Qu.    Median      Mean   3rd Qu. Max. NA's
## Logistic 0.7857143 0.8750000 0.9285714 0.9214286 0.9821429    1    0
## LASSO    0.8571429 0.8750000 0.9285714 0.9357143 1.0000000    1    0
## MARS     0.8571429 0.9285714 1.0000000 0.9642857 1.0000000    1    0
## LDA      0.8571429 0.9285714 1.0000000 0.9642857 1.0000000    1    0
##
## Spec
##               Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic 0.7692308 0.9230769 0.9230769 0.9010989 0.9271978 1.0000000    0
## LASSO    0.7692308 0.8461538 0.9230769 0.8857143 0.9271978 1.0000000    0
## MARS     0.7692308 0.8653846 0.9230769 0.9159341 0.9821429 1.0000000    0
## LDA      0.6923077 0.7884615 0.8461538 0.8401099 0.9065934 0.9285714    0
```

```r
bwplot(res, metric = "ROC", main = "CV ROC: Logistic vs. LASSO vs. MARS vs. LDA")
```



Based on 10-fold cross-validation, MARS is still the winning model.

```r
set.seed(1)
mars.probs <- predict(mars.fit,
                      newdata = testing_data,
                      type = "prob")[,2]

roc.mars <- roc(testing_data$mpg_cat, mars.probs)
```

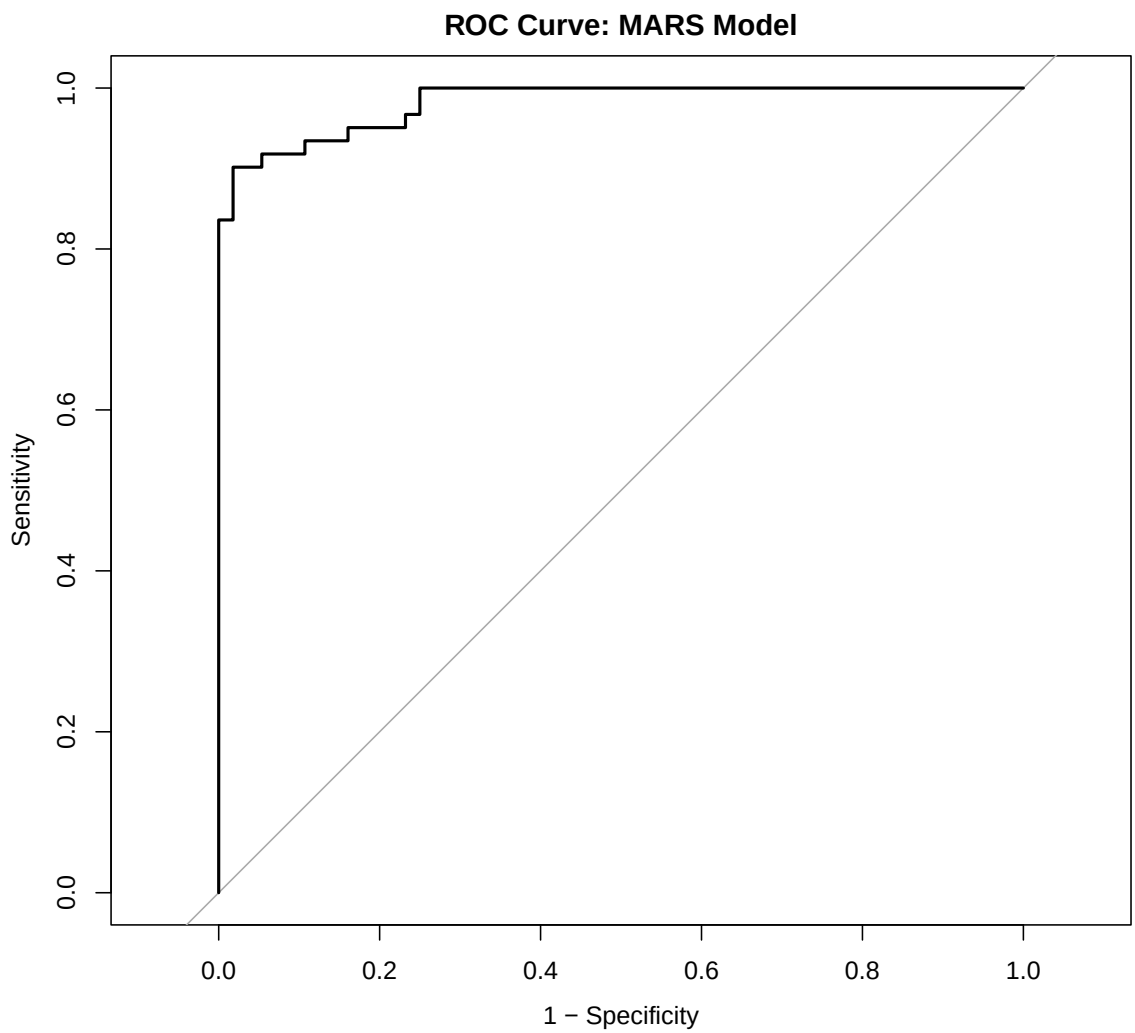
```
## Setting levels: control = high, case = low
```

```
## Setting direction: controls < cases
```

```r
auc(roc.mars)
```

```
## Area under the curve: 0.9816
```

```r
plot(roc.mars, legacy.axes = TRUE,
     main = "ROC Curve: MARS Model")
```



The AUC for MARS is 0.9816.

(e) Select a probability threshold to classify observations and compute the confusion matrix for the final model you choose. Briefly interpret what the confusion matrix indicates about your model's performance. (Note: We are looking for concise inter-pretations based on your own understanding. Do not use AI tools to answer this question.)

```r
test.probs <- predict(mars.fit, newdata = testing_data, type = "prob")
test.pred.prob <- test.probs[, "high"]

test.pred <- rep("low", length(test.pred.prob))
test.pred[test.pred.prob > 0.5] <- "high"

# confusion matrix
confusionMatrix(data = as.factor(test.pred),
                reference = testing_data$mpg_cat,
                positive = "high")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction high low
##       high   48   4
##       low     8  57
##
##                Accuracy : 0.8974
##                  95% CI : (0.8277, 0.9459)
##     No Information Rate : 0.5214
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.7939
##
##  Mcnemar's Test P-Value : 0.3865
##
##             Sensitivity : 0.8571
##             Specificity : 0.9344
##          Pos Pred Value : 0.9231
##          Neg Pred Value : 0.8769
##              Prevalence : 0.4786
##          Detection Rate : 0.4103
##    Detection Prevalence : 0.4444
##       Balanced Accuracy : 0.8958
##
##        'Positive' Class : high
##
```

I decided on a decision rule of 0.5 to classify between "High" and "Low" efficiency cars. This is equivalent to assuming that there are equal consequences associated with assigning False Positives (missclassification of a "Low"-efficiency car as "High") and False Negatives (missclassification of a "High"-efficiency car as "Low").

With an overall accuracy of 0.897, MARS will correctly classify the "High" efficiency vehicles 85.7% of the time (Sensitivity) and correctly classifies the "Low" efficiency vehicles 93.4% of the time.